MIL-C-39029/54B(USAF)
20 February 1981
SUPERSEDING
MIL-C-39029/54A(USAF)
13 October 1978

MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE, SHIELDED, (FOR MIL-C-26500(USAF) AND MIL-C-26518(USAF) CONNECTORS)

This specification sheet is approved for use by the Air Force Acquisition Logistics Division, Electronic Support Division (AFALD/PTES), Wright Patterson AFB, Ohio 45433, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the contacts described herein shall consist of this document and the latest issue of Specification MIL-C-39029.

INACTIVE FOR NEW DESIGN

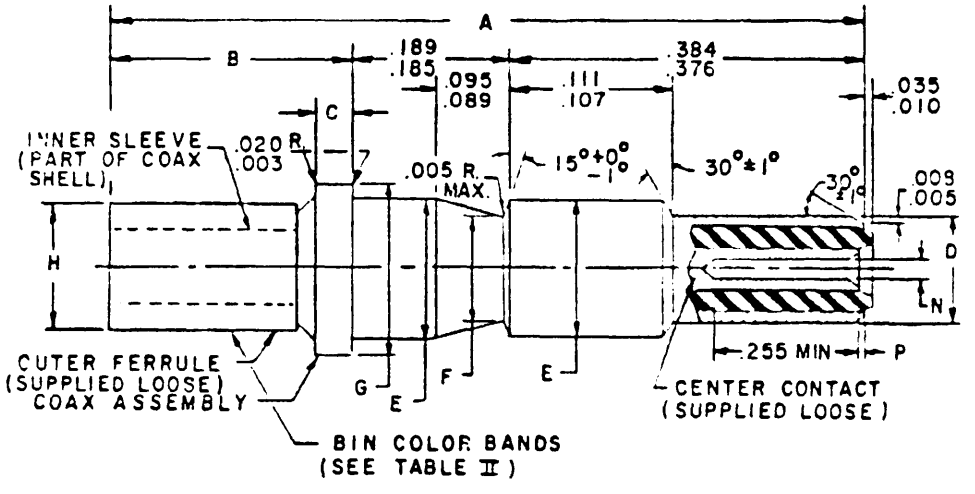


FIGURE 1. Connector contact.

Ⓑ denotes changes

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FSC 5935

Ⓑ TABLE I. Dimensions.

BIN code	A (REF)	B MAX	C	D DIA	E DIA	F DIA	G DIA	H MAX DIA	N DIA	P
342	.865 (21.97)	.298	.062	.124	.161	.135	.202 (5.13)	.154	.028	.066
			.044	.122	.158	.131	.196 (4.98)		.026	.020
343	.865 (21.97)	.298	.050	.215	.252	.226	.313 (7.95)	.195	.045	.016 Recessed
			.044	.213	.249	.221	.309 (7.85)		.043	.010 Protrusion

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Metric equivalents are in parentheses for overall length and diameter only.
4. Dimensions shown apply after plating.
5. Boot sealing: All dimensions are after curing, all diameters shall be concentric within .004 T.I.R. radii shall be .008 maximum unless otherwise specified.
6. Sleeve sealing: Color, white.

FIGURE 1. Connector contact - Continued.

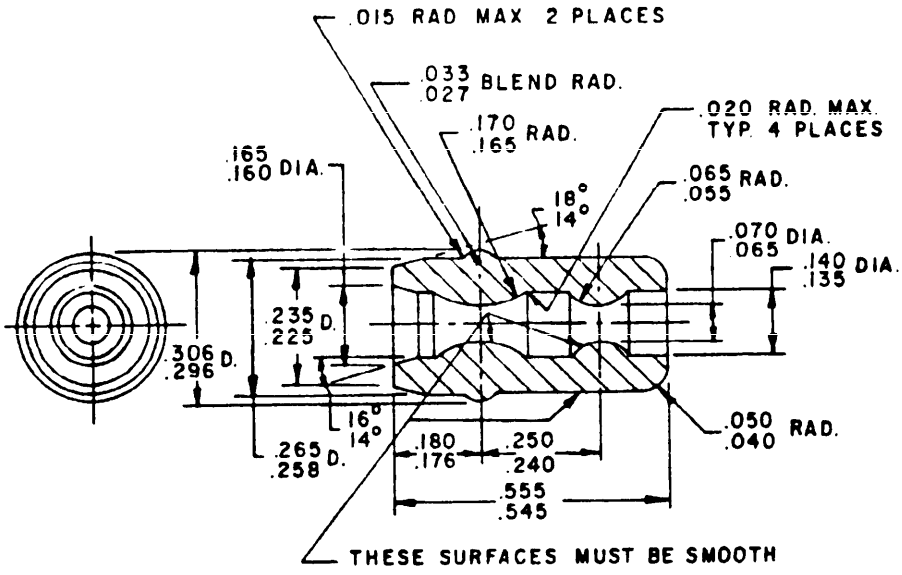


FIGURE 2. Boot, sealing.

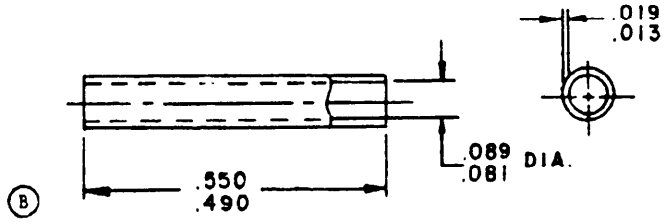


FIGURE 3. Sleeve, sealing, coaxial cable.

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BIN CODE 342 ASSEMBLY PROCEDURE (REF)

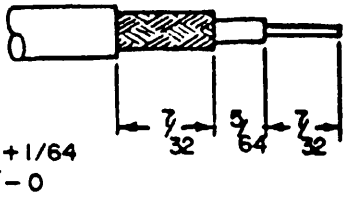
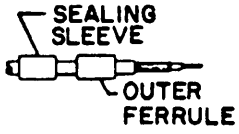
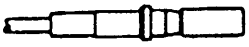
STEP 1 Strip cable jacket, braid and dielectric to the dimensions shown. Make all cuts square and sharp, being careful not to nick braid, dielectric, or center conductor when cutting. If conductor ends fray, twist them to their normal lay.	
STEP 2 After stripping, slide sealing sleeve and outer ferrule onto cable.	
STEP 3 Insert stripped center conductor into contact until wire shows through inspection hole and dielectric butts against contact. Fully seat contact in nest bushing of crimp tool and crimp in one full stroke.	
STEP 4 Slightly flare out ends of wire braid to facilitate insertion of inner ferrule of body assembly. (Do not comb braid.)	
STEP 5 Install center contact in body assembly and slide inner ferrule underneath wire braid as shown. Push center contact until it is locked in place in the body assembly. Pull lightly on cable to make sure that contact is securely locked in place.	
STEP 6 Slide outer ferrule over braid and up against body as shown. There should be no slack in the wire braid. Crimp the outer ferrule with proper tools. Then slide sealing sleeve toward contact until sleeve touches outer ferrule.	
STEP 7 Insert the assembled shielded contact into the connector in the same way as the standard contact with applicable insertion tool. This completes assembly.	

FIGURE 4. Assembly procedure.

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BIN CODE 343 ASSEMBLY PROCEDURE (REF)

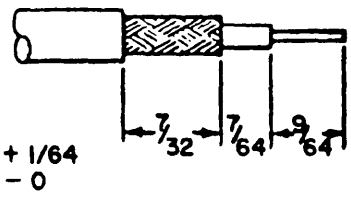
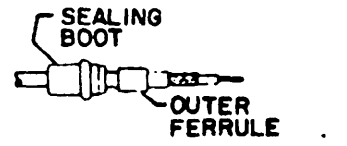
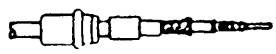
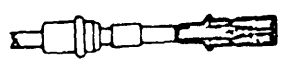
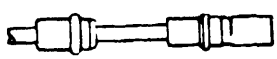
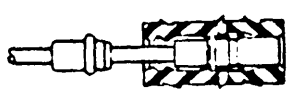
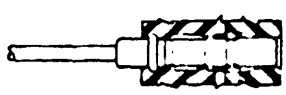
STEP 1 Strip cable jacket, braid and dielectric, to the dimensions shown. Make all cuts square and sharp, being careful not to nick braid, dielectric or center conductor when cutting. If conductor ends fray, twist them to their normal lay.	
STEP 2 After stripping, slide sealing boot and outer ferrule onto cable.	
STEP 3 Insert stripped center conductor into contact until wire shows through inspection hole and dielectric butts against contact. Fully seal contact in nest bushing of crimp tool and crimp in one full stroke.	
STEP 4 Slightly flare out ends of wire braid to facilitate insertion of inner ferrule of body assembly. (Do not comb braid)	
STEP 5 Install center contact in body assembly and slide inner ferrule underneath wire braid as shown. Push center contact until it is locked in place in the body assembly. Pull lightly on cable to make sure that contact is securely locked in place.	
STEP 6 Slide outer ferrule over braid and up against body as shown. There should be no slack in the wire braid. Crimp the outer ferrule with proper tools.	
STEP 7 Insert the assembled shielded contact into the connector in the same way as the standard contact with applicable insertion tool.	
STEP 8 To complete assembly, push sealing boot into connector grommet until O-ring riser of boot snaps into place and seals the assembly.	

FIGURE 4. Assembly procedure - Continued.

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TABLE II. Design characteristics.

BIN code	Color bands 1st	2nd	3rd	Contact cavity size	Ⓑ Cables accommodated	Type	Class
342	Orange	Yellow	Red	12	M27500-22RE1N06	D	A
343	Orange	Yellow	Orange	8	M27500-20RE1N06		

TABLE III. Tools.

BIN code	Inner contact: Basic crimping tool	Inner contact: Positioner	Outer contact: Basic crimping tool	Outer contact: Crimping die	Installing tool	Removal tool
342	M22520/2-01	M22520/2-33	M22520/5-01	M22520/5-08	M81969/17-05	M81969/19-09
			M22520/10-01	M22520/10-05(A)		
343	M22520/2-01	M22520/2-24	M22520/5-01	M22520/5-10	M81969/17-06	M81969/19-03
			M22520/10-01	M22520/10-06		

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1 thru 3 and table I.

Ⓑ Vibration: Method 2005 of MIL-STD-1344, test condition IV.

Ⓑ Shock (specified pulse): Method 2004 of MIL-STD-1344, test condition C.

Tools: See table III.

Boots and sleeve sealing: Parts must be free of flash, cracks or cavities (see figure 2).

Boots sealing: Immediately after molding, cure boots.

Sleeve sealing material: Flexite SR200 silicone rubber extruded tubing, or equivalent standard wall (see figure 3).

Mating contact: MIL-C-39029/55.

Assembly procedure: See figure 4.

Boots and sleeve sealing shall be shipped with units package.
Manufacturer's recommended assembly instructions to be shipped with unit package.

QPL evaluating activity: Defense Electronics Supply Center, (DESC-E), Dayton, OH 45444.

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Military part number: See table IV.

(B) TABLE IV. Military part number.

BIN code	Military part number	Supersedes part number
342	M39029/54-342	MS27184-22P
343	M39029/54-343	MS27184-20P

Custodian:
Air Force - 85

Review activities:
Air Force - 11, 17, 99
DLA - ES

Preparing activity:
Air Force - 85

(Project 5935-F160-1)

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